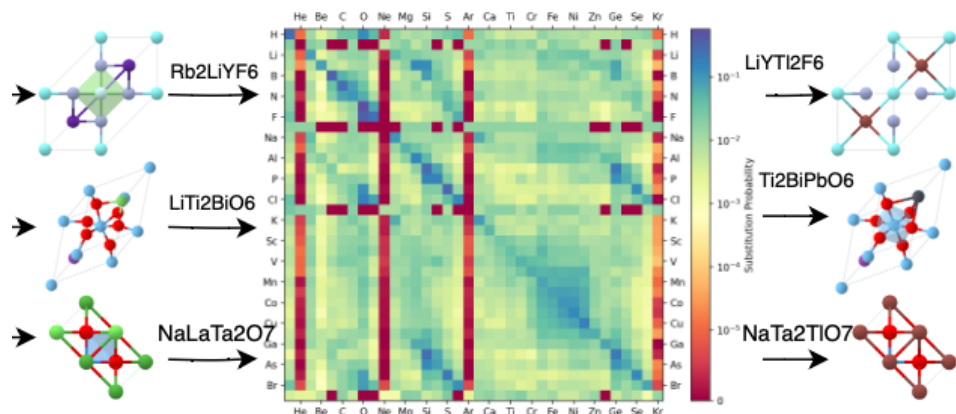
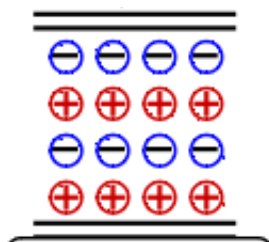


# Novel structure generation by atomic substitution



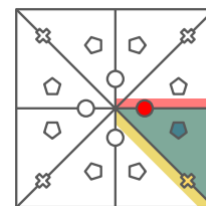
Experimental synthesis and testing



DFT calculations



Triage based on predicted figure of merit and stability



$$E_f = W_{ren} \left( \text{X} \quad \bullet \quad \text{pentagon} \right)$$

Coordinate-free machine learning prediction